



Module 1 : Introduction to NLP

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Agenda

- Basic Terms:
- NLP Preprocessing
- Tokenization
- stemming
- lemmatization



NLP Preprocessing – An Overview

- **NLP preprocessing** refers to the set of techniques used to **clean, normalize, and transform raw text data** into a structured format suitable for computational analysis.
- **Why Preprocessing is Required?** Natural language is:
 - Unstructured
 - Ambiguous
 - Inconsistent (case, spelling, grammar)

Preprocessing helps:

- Reduce noise
- Improve model accuracy
- Reduce vocabulary size
- Standardize text representation



NLP Preprocessing Steps

- Text normalization (lowercasing, punctuation removal)
- Tokenization
- Stop-word removal
- Stemming
- Lemmatization

Raw Text



Text Cleaning



Tokenization



Stopword Removal



Stemming / Lemmatization



Processed Text



Key Aspects of Text Cleaning

- **Normalization:** Converting all text to lowercase to ensure uniformity (e.g., "Data" and "data" become "data").
- **Noise Removal:** Removing HTML tags, URLs, email addresses, punctuation, and digits, other language script if any, particularly for social media data.
- **Stop Word Removal:** Eliminating common, low-value words (e.g., "and", "the") that do not add significant meaning.
- **Stemming/Lemmatization:** Reducing words to their root form (e.g., "running" to "run") to standardize vocabulary.
- **Handling Structure:** Removing extra spaces, converting encoding (like UTF-8), and handling contractions (eg.: do not - don't).



Text Processing : Tokenization

- Tokenization is the process of **breaking text into smaller units called tokens.**
- Tokens can be: **Words, Sentences, Characters, Sub-words**
- In **sentence segmentation**, the challenge is that where the sentence begin and end.
- **Few Challenges are:**
 - While '!', '?' are quite unambiguous.
 - Period '.' is quite **ambiguous** and can be used additionally for
 - Abbreviations (Dr., Mr., V.E.S.I.T)
 - Numbers (2.4%, 4.3)
- **Approach:** build a binary classifier, For each ".", need to decide End of Sentence / Not End Of Sentence



Word & Sentence Tokenization

- **Word Tokenization** : Splits text into individual words.
- **Example:** Input Text:
 - "NLP is transforming education."
 - Tokens:
 - ["NLP", "is", "transforming", "education"]
- **Sentence Tokenization** : Splits text into sentences
- **Example:** Input Text: "NLP is powerful. It is widely used."
- **Output Sentences:**
 1. NLP is powerful.
 2. It is widely used.

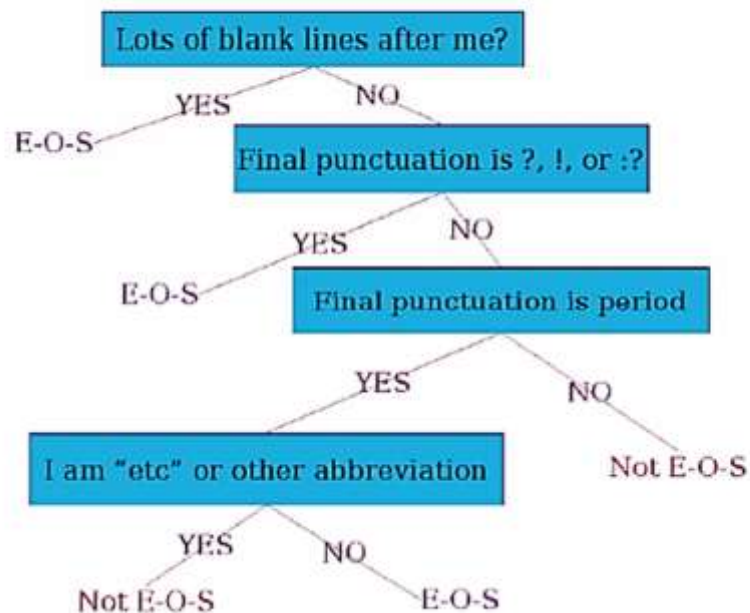


Word Tokenization

- Consider an Example:
I have a can opener; but I can't open these cans
- Word Token:
 - An occurrence of a word.
 - For the above sentence, 11 word tokens.
- Understand the query, understand the document, understand the data being communicated...

Sentence Segmentation : Example

- Decision Tree: Is this word the end-of-sentence (E-O-S)?





Character Tokenization

- Splits text into characters.
- Example
 - Input: "NLP"
 - Output: ['N', 'L', 'P']
- Sentence:
"Students are learning NLP"
- Tokenization:
| Students | are | learning | NLP |



Challenges in Tokenization

- Handling punctuation (don't, U.S.A)
- Indian languages with compound words
(Backbone - रीड की हड्डी reed kee haddee)
- Emojis and hashtags in social media
- Hyphenated words (state-of-the-art)



Applications of Tokenization

- Search engines
- Chatbots
- Text classification
- Machine translation

Stemming

- **Stemming** is the process of **reducing words to their root or base form** by removing suffixes.
- **Example:**

Original Word	Stem
playing	play
played	play
player	play
studies	studi
caring	car

1. Porter Stemmer
2. Snowball Stemmer
3. Lancaster Stemmer

In Hindi

- "ja" (जा) from "jaenge" (जाएंगे)
- किरणें (kirnen) (rays) -> किरण (kiran) (ray)



Stemming Example : Porter Stemmer

- **Example:**

Input Sentence: "Students are studying studies"

After Stemming: student are studi studi

Stemming Diagram

Words	Stemming
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playing → play

played → play

plays → play

Disadvantages of Stemming

- Produces incorrect roots
- Loss of meaning
- Not context-aware

Advantages of Stemming

- Faster processing
- Reduces vocabulary size
- Useful in Information Retrieval (IR)



Lemmatization

- **Lemmatization** reduces words to their **dictionary base form (lemma)** using:
- Vocabulary
- Morphological analysis
- Part-of-Speech (POS)
- The output is always a **valid word**.
- **Example:**

Word	POS	Lemma
running	verb	run
better	adjective	good
mice	noun	mouse
studies	noun	study



Lemmatization : Example

- **Input:** "The dogs were running fast."
- **Lemmatization:** "The dog be run fast."
(Here, *dogs* -> *dog*, *were* -> *be*, *running* -> *run*)
- **Input:** "He goes, she went, they go."
- **Lemmatization:** "He go, she go, they go." (All forms reduce to the base verb *go*)
- **Input (Devanagari):** "लड़के खेल रहे हैं।" (Ladke khel rahe hain.) - "Boys are playing."
- **Lemma (Root):** "लड़का खेलना।" (Ladka khelna.) - "Boy play (infinitive)."
- **Input (Devanagari):** "वह जाता है।" (Vah jaata hai.) - "He goes."
- **Lemma (Root):** "वह जाना।" (Vah jaana.) - "He go (infinitive)."
- **Key Differences from Stemming**
 - **Stemming (English):** *Caring* -> *Car* (often not a real word).
 - **Lemmatization (English):** *Caring* -> *Care* (a real dictionary word).



Lemmatization with Context

- **Word:** "saw"
- **Sentence 1:** "I saw a movie"
- **Lemma:** see
- **Sentence 2:** "I used a saw"
- **Lemma:** saw

Word + POS



Dictionary Lookup



Lemma (Valid Word)



Lemmatization vs Stemming

Feature	Stemming	Lemmatization
Speed	Fast	Slower
Accuracy	Low	High
Dictionary	No	Yes
Context	No	Yes
Output	May not be valid	Valid word



Lemmatization vs Stemming

- **Sentence:**
 - "Children are playing games"
- **Stemming:**
 - children are play game
- **Lemmatization:**
 - child be play game

Raw Text



Tokenization



Stemming / Lemmatization



Normalized Tokens



Lemmatization & Stemming: Exercises

Apply **stemming** and **lemmatization** to the sentence / word

- Students are learning technologies
- Why is lemmatization slower than stemming?
- Input: "Caring, cared, cares, care"
- Input: "Better, best"
- Researchers are analyzing datasets
- Teachers have corrected answers
- Students are preparing examinations
- The cars were moving faster
- Workers are building houses

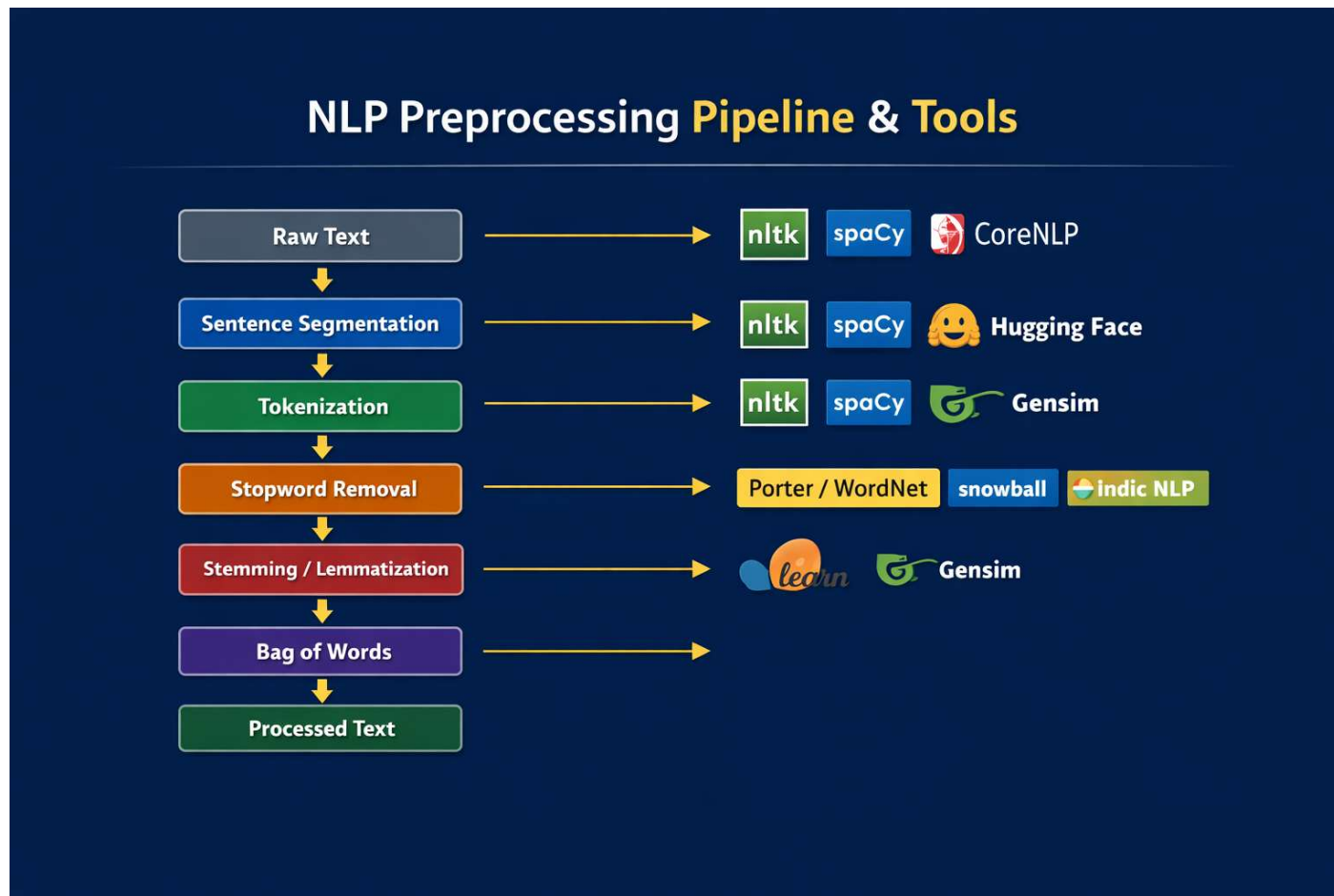


Lemmatization & Stemming: Exercises

Solution (English)

1. Stemming output: student are learn technolog and Lemmatization output: student be learn technology
2. Lemmatization is slower because it uses dictionaries, part-of-speech tagging, and morphological analysis
3. Stemming: Car, car, car, car (Porter/Simple) and Lemmatization: Care, care, care, care (Uses context, dictionary)
4. Stemming: Better, best (Porter, Snowball), bet, best (Lancaster stem) and Lemmatization: Good, best (Uses POS for 'better' as adverb/adj)
5. Stemming: research are analyz dataset and Lemma: researcher be analyze dataset
6. teacher have correct answer and LEMma: teacher have correct answer
7. student are prepar examin and LEMma: student be prepare examination
8. the car were move faster and Lemma: the car be move fast
9. worker are build hous and LEMma: worker be build house

NLP Preprocessing Pipeline & Tools





Stemming: Tools Used

1. NLTK Porter Stemmer

 <https://www.nltk.org/api/nltk.stem.porter.html>

2. NLTK Snowball Stemmer

 <https://www.nltk.org/api/nltk.stem.snowball.html>

3. Lancaster Stemmer (NLTK)

 <https://www.nltk.org/api/nltk.stem.lancaster.html>

4. Indic NLP Stemmer (Hindi, Marathi)

 https://github.com/anoopkunchukuttan/indic_nlp_library



Lemmatization: Tools Used

1. NLTK WordNet Lemmatizer

 <https://www.nltk.org/api/nltk.stem.wordnet.html>

2. spaCy Lemmatizer (POS-based)

 <https://spacy.io/usage/linguistic-features#lemmatization>

3. Stanford CoreNLP Lemmatizer

 <https://stanfordnlp.github.io/CoreNLP/lemma.html>

4. Indic NLP Lemmatization Resources

 <https://ltrc.iiit.ac.in/showfile.php?filename=downloads/nlp-tools>